Deep Learning-Based Facial Expression Recognition Using Convolutional Neural Networks

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Contents

Abstract…… 3

Introduction……..……4

Data Description……...5-6

Methodology…….7-9

Data Preprocessing……10-11

Model……12-15

Conclusion……16

Abstract

Facial emotion recognition is an important application of computer vision and deep learning that aims to identify emotional expressions from facial images. This project develops a multiclass image classification model using transfer learning with ResNet50, a convolutional neural network pretrained on the ImageNet dataset. The model classifies facial images into eight emotion categories: anger, contempt, disgust, fear, happy, neutral, sad, and surprise.

The project involves image preprocessing, data augmentation, model training, fine-tuning, and performance evaluation. Images are resized to (224 x 224) pixels and processed in batches of 32. Random horizontal flipping, rotation, and zooming are used to improve the model's ability to generalize to variations in facial images. The pretrained ResNet50 architecture is combined with a custom classification head to perform emotion classification.

The initial evaluation reported an accuracy of approximately 58%, with a weighted F1-score of 0.56. The model performed relatively well in identifying happy and neutral expressions, while disgust and fear were more difficult to classify. These findings demonstrate the potential of transfer learning for facial emotion recognition while highlighting the challenges associated with distinguishing visually similar expressions and achieving consistent performance across all emotion categories.

Introduction

Facial expressions are an important component of human communication. People often use facial movements to convey emotions such as happiness, sadness, anger, fear, and surprise. Automatically recognizing these expressions using computer vision has become an important research area in artificial intelligence and human–computer interaction.

Facial Emotion Recognition (FER) is the process of identifying and classifying facial expressions into predefined emotion categories using digital images or video frames. Deep learning techniques, particularly Convolutional Neural Networks (CNNs), have been widely used for image classification because they can learn relevant visual features directly from image data.

Traditional image classification methods often depend on manually designed features. In contrast, deep learning models learn hierarchical representations of images, ranging from basic visual patterns to more complex facial features. However, facial emotion recognition remains challenging because expressions can vary across individuals, lighting conditions, head orientations, image quality, and expression intensity.

The objective of this project is to develop a deep learning model capable of classifying facial images into eight emotion categories. The project uses transfer learning with ResNet50 to leverage features learned from a large image dataset and adapt them to facial emotion classification.

The main objectives of the project are:

- To load and preprocess facial images for deep learning.
- To explore the available dataset and its emotion categories.
- To apply data augmentation to introduce variation into the training images.
- To develop a ResNet50-based multiclass classification model.
- To train the model using transfer learning and fine-tuning.
- To evaluate model performance using accuracy, precision, recall, F1-score, and a confusion matrix.
- To identify the limitations of the model and suggest possible improvements.

The project demonstrates an end-to-end deep learning workflow, from image preparation to model evaluation.

Data Description

The project uses an image dataset organized into separate training and testing directories. Each directory contains subfolders corresponding to the different emotion classes. The labels are automatically assigned based on the directory structure using TensorFlow/Keras image-loading utilities.

The dataset contains eight emotion categories: Anger, Contempt, disgust, fear, happy, neutral, sad, and surprise.

The dataset is loaded from a compressed archive stored in Google Drive and extracted into the Google Colab environment. The images are subsequently loaded and resized to a fixed dimension suitable for the ResNet50 architecture.

Dataset summary

Parameter	**Description**
Total training images	16,108
Total test images	14,518
Total images	30,626
Number of emotion classes	8
Image dimensions	(224 x 224) pixels
Number of colour channels	3
Batch size	32
Task	Multiclass image classification
Deep learning framework	TensorFlow/Keras
Model architecture	ResNet50

Emotion categories

The target variable represents the emotion category associated with each facial image.

Emotion	Description
Anger	Facial expressions associated with anger
Contempt	Facial expressions associated with contempt
Disgust	Facial expressions associated with disgust
Fear	Facial expressions associated with fear
Happy	Facial expressions associated with happiness
Neutral	Facial expressions without a clearly expressed target emotion
Sad	Facial expressions associated with sadness
Surprise	Facial expressions associated with surprise

The number of images varies across the emotion categories. In the test dataset, happy contains 2,704 images, whereas disgust contains 1,248 images. This variation in class frequencies is relevant when interpreting the model's performance.

The dataset is suitable for supervised learning because each image is associated with a predefined class label. However, the original data source, collection procedure, demographic composition, and licensing details are not known

Methodology

The methodology consists of several stages: image loading, preprocessing, data augmentation, model construction, training, fine-tuning, and performance evaluation.

Image loading and preprocessing

The dataset is loaded using the tf.keras.utils.image_dataset_from_directory() function. This function reads images from the class-specific folders and automatically assigns integer labels to the corresponding emotion categories.

All images are resized to (224 x 224) pixels to ensure a consistent input dimension. The images are processed in batches of 32 to facilitate efficient training and reduce memory consumption.

The ResNet50-specific preprocess_input() function is applied to transform the image pixel values into the format expected by the pretrained network. This preprocessing step is important because the model uses weights learned from ImageNet.

The test dataset is loaded without shuffling so that the predicted labels remain aligned with the corresponding true labels during evaluation.

Data augmentation

Data augmentation is used to increase the variability of the training images and potentially reduce overfitting. The following augmentation techniques are applied:

- Random horizontal flipping: Produces horizontally flipped versions of images to introduce variation in facial orientation.
- Random rotation: Rotates images slightly to help the model handle changes in facial orientation.
- Random zooming: Changes the apparent scale of facial images to introduce variation in framing.

These transformations are applied during training. The purpose is to help the model learn more generalizable visual features rather than relying too heavily on specific image characteristics.

ResNet50 architecture

ResNet50 is a deep convolutional neural network that uses residual connections to facilitate the training of deeper networks. It is pretrained on ImageNet and used as the feature extraction component of the proposed model.

Instead of training the entire network from scratch, the original classification head is removed and replaced with a custom classification head designed for the eight emotion categories.

The architecture consists of the following components:

1. ResNet50 base: Extracts visual features from the input facial images.
2. Global Average Pooling: Converts the extracted feature maps into a compact feature vector.
3. Dropout layer (0.3): Randomly deactivates a proportion of units during training to help reduce overfitting.
4. Dense layer (128 neurons): Learns task-specific representations using the ReLU activation function.
5. Dropout layer (0.2): Provides additional regularization.
6. Output layer (8 neurons): Uses the softmax activation function to estimate the probability of each emotion category.

The softmax function is defined as

$$P(y = i \mid x) = \frac{e^{z_i}}{\sum_{j=1}^{8} e^{z_j}},$$

where z_i is the output score for class i. The predicted emotion is the category with the highest estimated probability.

Model training

The model is trained using a two-stage transfer learning approach.

Stage 1: Feature extraction

Initially, the pretrained ResNet50 layers are frozen, meaning their weights are not updated during training. Only the newly added classification layers are trained.

The Adam optimizer and sparse categorical cross-entropy loss function are used. Early stopping monitors validation accuracy and stops training if there is no improvement for three consecutive epochs, restoring the best weights.

Stage 2: Fine-tuning

After the initial training stage, the model is fine-tuned by making the backbone trainable while unfreezing only its final 20 layers. Earlier layers remain frozen, and batch normalization layers are kept frozen to reduce training instability.

The model is recompiled using the Adam optimizer with a learning rate of 10^{-5}. The ReduceLROnPlateau callback reduces the learning rate when validation loss stops improving.

Fine-tuning allows the final layers of the pretrained network to adapt their learned representations to facial emotion recognition.

Class weighting

The notebook also implements an additional training run using class weights to account for differences in class frequencies.

Class weighting assigns different importance to observations belonging to different classes. This can help a model pay greater attention to underrepresented classes.

Performance evaluation

The model is evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

These metrics provide information about both overall predictive performance and the model's ability to recognize individual emotion categories.

An important consideration is that the notebook uses the test dataset as validation data during training and subsequently evaluates the model on the same dataset. Therefore, the reported performance should not be treated as an independent final test estimate.

Data Preprocessing

Dataset size

The dataset contains 16,108 training images and 14,518 test images, giving a total of 30,626 images across both directories.

The training dataset provides examples from which the model learns facial features, while the test directory is intended to represent unseen examples. However, because the test directory is also used during model training for validation, it does not serve as a fully independent test set in the current implementation.

Class distribution

The test-set distribution across the eight emotion categories is presented below.

Emotion	Number of test images
Anger	1,718
Contempt	1,312
Disgust	1,248
Fear	1,664
Happy	2,704
Neutral	2,368
Sad	1,584
Surprise	1,920
Total	14,518

The distribution shows that the number of test images differs across emotion categories. Happy has the largest number of test images, whereas disgust has the smallest.

This variation should be considered when evaluating classification performance. Accuracy alone may not adequately reflect the model's ability to recognize less

frequent classes. Macro-averaged metrics and class-wise recall provide additional information about performance across categories.

Image dimensions and preprocessing

The original images are resized to a uniform dimension of (224 x 224) pixels with three colour channels. Standardizing image dimensions allows the images to be processed in batches by the same neural network architecture.

The notebook does not provide a detailed analysis of the original image dimensions, brightness, contrast, image corruption, or duplicate images. Consequently, no specific conclusions can be drawn about these characteristics.

Importance of data augmentation

The use of horizontal flipping, rotation, and zooming introduces variation into the training images. This is particularly relevant to facial image classification because the appearance of a face may change with orientation, framing, and other visual conditions.

However, augmentation does not guarantee improved performance. Its effectiveness depends on whether the transformations preserve meaningful facial information and resemble variations that may occur in real-world images.

Model

Model performance evaluation determines how effectively the trained classifier identifies the correct emotion categories. Because this is a multiclass classification problem, several evaluation metrics are used to obtain a more complete understanding of performance.

Evaluation metrics

Accuracy

Accuracy measures the proportion of correctly classified images among all evaluated images.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

A higher accuracy indicates that the model correctly classifies a larger proportion of the images.

Precision

Precision measures the proportion of images predicted to belong to a particular class that actually belong to that class.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Here, TP represents true positives and FP represents false positives.

Recall

Recall measures the proportion of actual examples of a class that the model correctly identifies.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Here, FN represents false negatives.

F1-score

The F1-score combines precision and recall using their harmonic mean.

$$F_1 = 2\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The F1-score is particularly useful when precision and recall differ substantially.

Confusion matrix

A confusion matrix compares the actual emotion labels with the predicted labels. It helps identify which categories are frequently confused with one another.

Classification results

The initial classification report generated by the notebook provides the following results.

Emotion	Precision	Recall	F1-score	Support
Anger	0.57	0.31	0.40	1,718
Contempt	0.54	0.59	0.56	1,312
Disgust	0.53	0.15	0.23	1,248
Fear	0.59	0.29	0.39	1,664
Happy	0.86	0.86	0.86	2,704
Neutral	0.69	0.90	0.78	2,368
Sad	0.35	0.70	0.47	1,584
Surprise	0.41	0.45	0.43	1,920
Macro average	—	—	0.52	14,518
Weighted average	—	—	0.56	14,518

The overall accuracy reported in the notebook is approximately 58%.

Interpretation of results

The results indicate that the model performs differently across the eight emotion categories.

Happy: The model achieves a precision of 0.86, recall of 0.86, and F1-score of 0.86. This is the strongest class-wise result, indicating that happy expressions are recognized more consistently than the other categories.

Neutral: The model achieves a recall of 0.90 and an F1-score of 0.78. The high recall indicates that the model correctly identifies a large proportion of actual

neutral examples, although its precision of 0.69 suggests that some images predicted as neutral belong to other categories.

Contempt: The precision of 0.54 and recall of 0.59 indicate moderate performance. The model identifies a reasonable proportion of actual contempt examples, but confusion with other categories remains possible.

Sad: The model achieves a recall of 0.70 but a precision of only 0.35. This means that many actual sad examples are identified, but a substantial proportion of images predicted as sad belong to other categories.

Anger and fear: The recall values are 0.31 and 0.29, respectively. These results indicate that the model misses many actual examples from these categories.

Disgust: The model has the lowest recall, at 0.15, and an F1-score of 0.23. This indicates substantial difficulty identifying disgust expressions correctly.

Surprise: The model achieves a precision of 0.41, recall of 0.45, and F1-score of 0.43, suggesting moderate but inconsistent classification performance.

Overall, the difference between the macro F1-score of 0.52 and weighted F1-score of 0.56 indicates that the model's performance is uneven across classes. The weighted average gives greater influence to classes with more observations, whereas the macro average gives equal weight to every emotion category.

Fine-tuning performance

During fine-tuning, the notebook reports validation accuracy increasing from approximately 0.50 in the early stage to around 0.58 in later logged epochs.

Training accuracy increases more than validation accuracy, while validation loss fluctuates around 1.4–1.5. This pattern suggests that further improvements in training performance do not consistently translate into better validation performance.

After the additional class-weighted training run, the notebook reports the following overall metrics:

Metric	Initial evaluation	After class-weighted run
Accuracy	0.58	0.57
Macro F1-score	0.52	0.51

Weighted F1-score	0.56	0.55

The reported results do not show an improvement in the overall metrics after the additional class-weighted training run. Furthermore, because the weights were calculated using test labels, the additional run requires a corrected experimental design before its results can be considered reliable.

Overall assessment

The ResNet50-based model demonstrates that transfer learning can be applied successfully to facial emotion classification. However, an accuracy of approximately 58% and uneven class-wise F1-scores indicate that the current model has substantial room for improvement.

Future development should focus on improving recognition of disgust, fear, and anger, investigating class-specific errors, and evaluating the model using a separate validation set and an untouched test set. The confusion matrix generated by the notebook should also be included in the final project submission to illustrate the specific patterns of misclassification.

Conclusion

his project developed a deep learning model for facial emotion recognition using transfer learning with ResNet50. The objective was to classify facial images into eight emotion categories: anger, contempt, disgust, fear, happy, neutral, sad, and surprise.

The workflow involved image loading, resizing, preprocessing, data augmentation, model construction, two-stage training, fine-tuning, and performance evaluation. The pretrained ResNet50 network was combined with a custom classification head consisting of global average pooling, dropout layers, a dense hidden layer, and a softmax output layer.

The initial evaluation reported approximately 58% accuracy, a macro F1-score of 0.52, and a weighted F1-score of 0.56. The model achieved its strongest results for happy and neutral expressions, whereas disgust, fear, and anger were more difficult to recognize. The additional class-weighted training run did not improve the reported overall metrics.

The findings demonstrate the usefulness of transfer learning as a starting point for facial emotion recognition while highlighting the importance of class-wise evaluation, careful data preparation, and appropriate validation procedures.

Future improvements should include a separate validation set, class-weight calculation using training data only, more comprehensive exploratory analysis, and systematic investigation of classification errors. The model should be considered an experimental facial-expression classifier rather than a definitive method for determining a person's actual emotional state.